

Multiple choice 2020 - HSC - 2 unit Task 2 (Answers)

Q.1. $y = \sin\left(x \times \frac{\pi}{180}\right)$
 $\frac{dy}{dx} = \frac{\pi}{180} \times \cos \frac{\pi}{180} x$

(C)

Q.2. $\lim_{x \rightarrow 0} \frac{\sin 2x}{3x} = \frac{2}{3}$

(A)

Q.3



$$\int_0^3 f(x) dx = 6$$

$$\int_0^3 f(x) dx = 5 \therefore \int_1^3 f(x) dx = -1$$

(A)

Q.4 $y = 4 \sin\left(\frac{x}{3}\right)$ $T = \frac{2\pi}{\frac{1}{3}} = 6\pi$

(A)

Q.5 $y = \log_e(2x-1) \therefore 2x-1 > 0$
 $x > \frac{1}{2}$

(C)

Q.6 $\int_0^1 (x-1)^2 - (2x-2) dx + \int_1^3 (2x-2) - (x-1)^2 dx$

(D)

Q.7 $f(x) = \ln(\tan x)$

$$f'(x) = \frac{\sec^2 x \rightarrow f'(x)}{\tan x \rightarrow f(x)} = \frac{\frac{1}{\cos^2 x}}{\frac{\sin x}{\cos x}} = \frac{1}{\cos x} \cdot \frac{1}{\sin x} = \sec x \cdot \operatorname{cosec} x$$

(C)

Q.8 $\int_1^d \frac{2}{x} dx = 2 \therefore 2[\ln|x|]_1^d = 2$
 $d > 0$ $\ln d - \ln 1 = 1 \therefore \ln d = 1 \therefore e^1 = d$

(A)

Q.9 $f(x) < 0 \therefore A, C, D$

$f'(x) > 0 \therefore$ increasing $\therefore A, D$

$f''(x) < 0 \therefore$ concave down

$\therefore$ (A)

Q.10 $y = x^2 + bx + c$, $x = \frac{-b}{2a}$ axis passes through vertex which is min. at $x = -2$
 $\therefore -2 = \frac{-b}{2 \times 1} \therefore b = 4$ since $f'(-2) = 0$

now: $y = x^2 + 4x + c$ min. value $(-2, 6)$

$6 = (-2)^2 + 4(-2) + c \therefore c = 10$

(D)

Section II- Extended Response

Attempt Questions 11-16.

Allow about 75 minutes for this section.

Question 11(12 Marks)

a) Differentiate the following

(i) $y = \log_e(2x - 1)^4$

$$y = 4 \log_e(2x - 1)$$

$$\frac{dy}{dx} = 4 \times \frac{2}{2x-1}$$

$$\frac{dy}{dx} = \frac{8}{2x-1}$$

Marking
scale

(2 marks) 2

• correct solus. - (2)

• applies correctly

$\frac{d}{dx} \ln f(x) - (1)$

(ii) $y = \frac{x}{\sin^2 x}$

2

$$u = x \quad u' = 1$$

$$v = \sin^2 x \quad v' = 2 \sin x \cos x$$

$$\therefore \frac{dy}{dx} = \frac{1 \times \sin^2 x - 2x \sin x \cos x}{\sin^4 x}$$

or simplifies further $\frac{dy}{dx} = \frac{\sin x - 2x \cos x}{\sin^3 x}$

(2 marks)

• correct solus - (2)

• applies quotient rule correctly - (1)

(iii) $y = 3^x$

1

$$\frac{dy}{dx} = 3^x \ln 3$$

(1 mark)

• correct solus.

Question 11 continued on the next page

b) Integrate the following

$$\int 3e^{5x+1} - \frac{1}{x^3} dx$$

2

$$= \frac{3}{5} e^{5x+1} - \frac{x^{-2}}{-2} + C$$

(2 marks)

• correct solns.

$$= \frac{3}{5} e^{5x+1} + \frac{1}{2x^2} + C$$

(1 mark)

• Finds the correct primitive of $3e^{5x+1}$

(1 mark)

• Finds the correct primitive of $\frac{1}{x^2}$

c) Evaluate $\int_0^{\frac{\pi}{8}} \sec^2 2x dx$

2

$$= \frac{1}{2} \left[\tan 2x \right]_0^{\pi/8}$$

(2 marks)

• correct solns.

(1 mark)

$$= \frac{1}{2} \left[\tan \frac{\pi}{4} - \tan 0 \right]$$

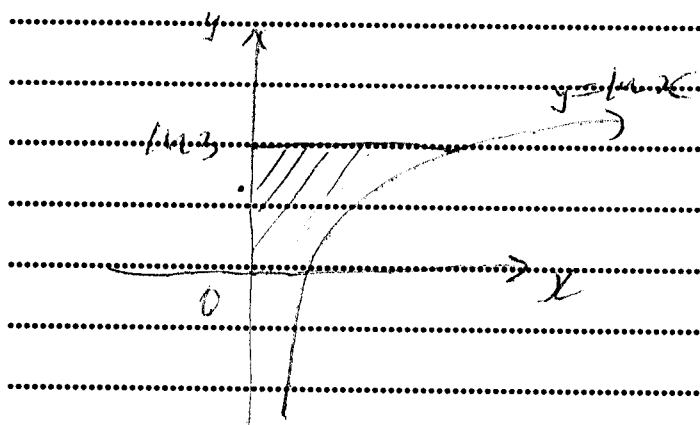
• Find the correct primitive

$$= \frac{1}{2} [1 - 0] = \frac{1}{2}$$

• correctly evaluates definite integral

Question 11 continued on the next page

- d) Find the area bounded by $y = \ln x$, y -axis between $y = 0$ and $y = \ln 3$. 3



(3 marks)

• correct solus.

(2 marks)

• correctly makes x -the subject and establishes the area towards y -axis

$$\text{Area} = \int x \, dy$$

where $y = \ln x$
 $x = e^y$

$$\therefore \text{Area} = \int_0^{\ln 3} e^y \, dy$$

$$= \left[e^y \right]_0^{\ln 3}$$

$$= e^{\ln 3} - e^0$$

$$= 3 - 1 = 2 \text{ (units}^2\text{)}$$

(1 mark)

• correctly makes x -the subject
 • finds the correct primitive

End of Question 11

Question 12 (7 Marks)

(a)

(i) Differentiate $\cos 3x^3$

→ (2)

$$\frac{d}{dx} \cos 3x^3 = 9x^2 (-\sin 3x^3)$$

(2 marks)

• correct solus

$$\therefore \frac{d}{dx} \cos 3x^3 = -9x^2 \sin(3x^3)$$

(1 mark)

• applies chain rule correctly

• correctly differentiates cos-function

(ii) Hence $\int x^2 \sin 3x^3 dx$

2

from (i) $\frac{d}{dx} \cos 3x^3 = -9x^2 \sin 3x^3$

(2 marks)

• correct solus

$$\therefore \cos 3x^3 = \int -9x^2 \sin 3x^3 dx$$

(1 mark)

$$\cos 3x^3 = -9 \int x^2 \sin 3x^3 dx$$

• correctly applies reverse chain rule

$$\therefore \frac{1}{-9} \cos 3x^3 = \int x^2 \sin 3x^3 dx$$

• ignore $+C$

Question 12 continued on the next page

(b) Solve the equation given by $2\ln x = \ln(3 - 2x)$

(3)

$$2 \ln x = \ln(3 - 2x)$$

(3 marks)

correct solns.

$$\ln x^2 = \ln(3 - 2x)$$

(2 marks)

$$x^2 = 3 - 2x$$

• gets both solns.

$$x^2 + 2x - 3 = 0$$

correctly but

$$(x+3)(x-1) = 0$$

doesn't exclude

$$x = -3$$

$$x = -3 \text{ or } x = 1$$

(1 mark)

but $x = -3$ is not a soln. • applies log. rules

since $2\ln(-3)$ is not defined

correctly

$$\therefore \text{soln: } (x=1)$$

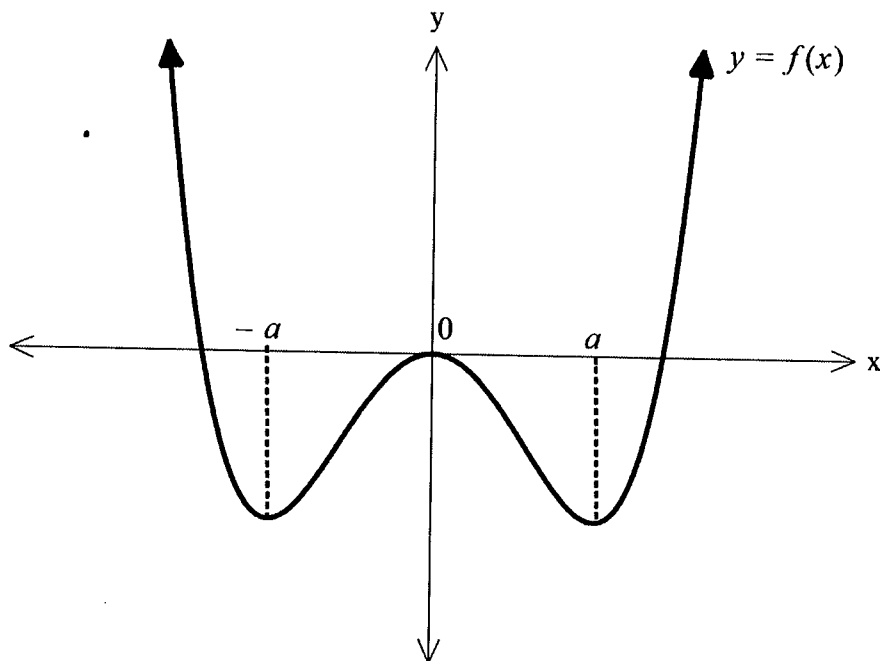
• solves quadratic equation correctly

End of Question 12

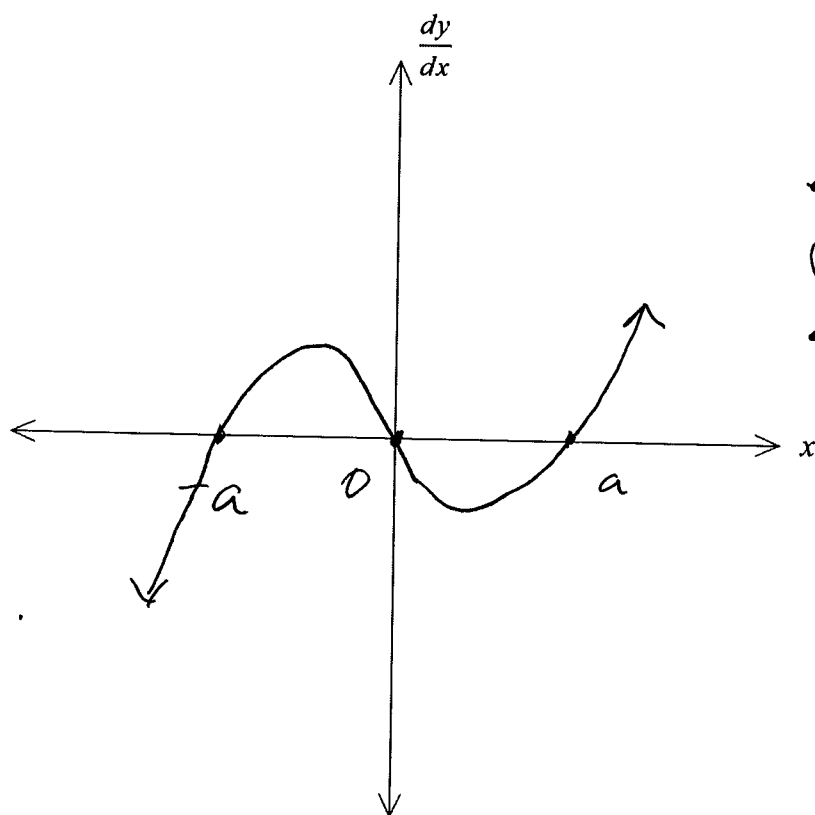
Question 13 (5 Marks)

(a) Given the graph of $y = f(x)$ below

2



Sketch a possible graph of $\frac{dy}{dx}$ on the axes provided.



2 marks

• correct graph

1 mark

• All 3 x-inter.
correct

• correct shape

Question 13 continued on the next page

- (b) Water is flowing out of a pool at a rate given by $R = 4t - t^2$ cubic metres/minute. 3
If the volume of water in the pool is initially 10 cubic metres, find the amount of the water in the pool after 4 minutes.

$$R = \frac{dV}{dt}$$

3 marks

• correct solus.

$$\therefore V = \int (4t - t^2) dt$$

2 marks

$$V = -2t^2 + \frac{t^3}{3} + C$$

• finds the correct primitive and evaluates the constant using the initial values

given $t = 0$ $V = 10 \text{ m}^3$

$$\therefore 10 = 2(0) - \frac{0}{3} + C$$

$$\therefore C = 10$$

1 mark

$$\therefore V = -2t^2 + \frac{t^3}{3} + 10$$

• finds the correct primitive
• finds Volumes correctly

Now after $t = 4$ minutes
sub. $t = 4$

$$V = -2(4)^2 + \frac{4^3}{3} + 10$$

$$V = \frac{62}{3} \text{ m}^3 \text{ or } = 20.6 \text{ m}^3$$

$\therefore$ pool is empty

OR

End of Question 13

Question 14 (8 marks)

For the curve given by $f(x) = 2xe^{-x}$

- (i) Show that $f'(x) = 2e^{-x}(1-x)$.

1

$$\begin{aligned}
 f(x) &= 2xe^{-x} \\
 u &= 2x & v &= e^{-x} \\
 u' &= 2 & v' &= -e^{-x} \\
 \therefore f'(x) &= 2e^{-x} + 2x(-e^{-x}) \\
 &= 2e^{-x} - 2xe^{-x} \\
 &= 2e^{-x}(1-x)
 \end{aligned}$$

1 mark
• correct solns.

- (ii) Hence or otherwise find any stationary point(s) and determine their nature.

2

$$\begin{aligned}
 f'(x) &= 0 \\
 2e^{-x}(1-x) &= 0 \\
 2e^{-x} &\neq 0 & 1-x &= 0 \\
 & & x &= 1
 \end{aligned}$$

x	0	1	2
f'(x)	2	0	-0.27

+ / - : (1, 2e⁻¹) Max. turning pt.

2 marks
• correct solns.
1 mark
• solves correctly
• justifies the nature of stat. pt. correctly

- (iii) Determine $\lim_{x \rightarrow \infty} f(x)$.

1

$$\lim_{x \rightarrow \infty} 2xe^{-x} = 0$$

1 mark
• correct solns.

Question 14 continued on the next page

(iv) Find any possible point of inflection.

$$f'(x) = 2e^{-x} - 2xe^{-x}$$

$$f''(x) = -2e^{-x} - (2e^{-x} - 2xe^{-x})$$

2
2 marks

• correct solns.

1 mark

$$\therefore f''(x) = -2e^{-x} - 2e^{-x} + 2xe^{-x}$$

$$= -4e^{-x} + 2xe^{-x}$$

$$= -2e^{-x}(2 - x)$$

• solves correctly
 $f''(x) = 0$

• shows change
in concavity

$$0 = -2e^{-x}(2 - x) \quad \text{at } x = +2$$

$$-2e^{-x} \neq 0 \quad x = +2 \quad \text{possible pt. of inflex.}$$

x	1	+2	3
f''	0.7	0	0.099

Change in concavity

$\therefore (2, 4e^{-2})$ pt. of inflexion

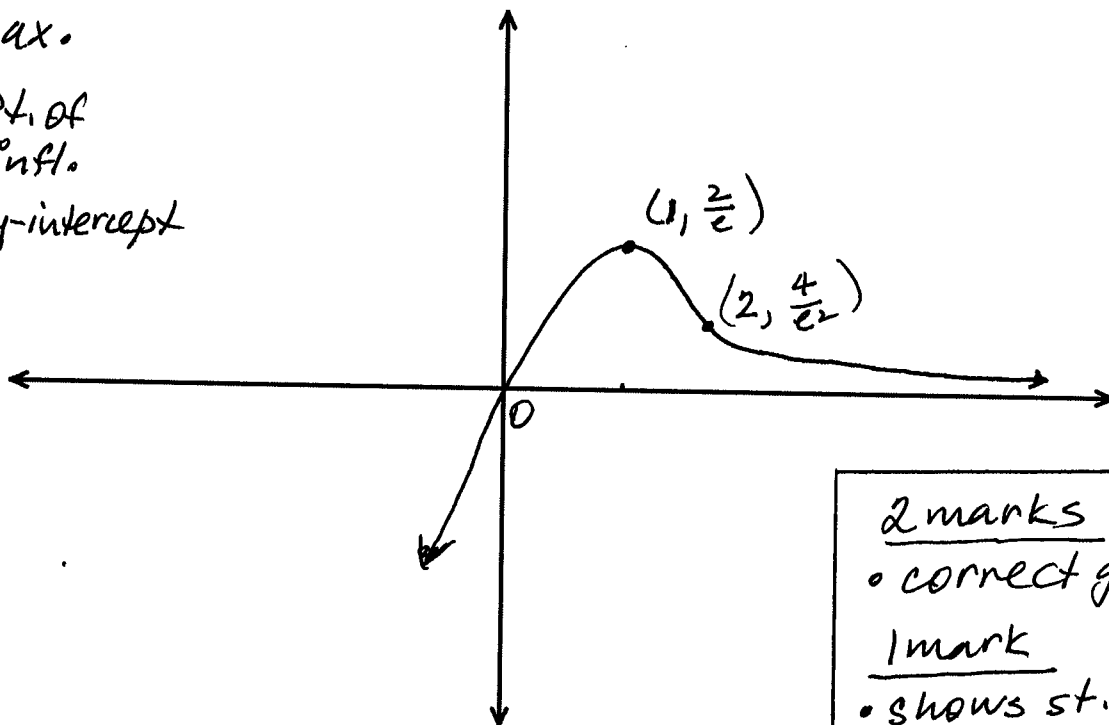
(v) Hence sketch the curve.

2

$(1, \frac{2}{e})$ Max.

$(2, \frac{4}{e^2})$ Pt. of inflex.

$(0, 0)$ - x, y-intercept



2 marks

• correct graph

1 mark

• shows st. point
& x-intercept

• correct shape
showing
asymptote

End of Question 14

Question 15 (10 marks)

The velocity of the particle is given by $v = 2 \sin\left(t - \frac{\pi}{3}\right)$ metres per second.

a) Find:

- (i) The equation of the displacement x of the particle as a function of time if the particle is initially at the origin.

2

$$x = \int v dt$$

$$x = \int 2 \sin\left(t - \frac{\pi}{3}\right) dt$$

$$x = -2 \cos\left(t - \frac{\pi}{3}\right) + C$$

$$\text{at } t=0 \quad x=0$$

$$\therefore 0 = -2 \cos\left(0 - \frac{\pi}{3}\right) + C$$

$$C = 1$$

$$\therefore x = -2 \cos\left(t - \frac{\pi}{3}\right) + 1$$

2 marks

• correct solns.

1 mark

• correctly finds primitive of $2 \sin\left(t - \frac{\pi}{3}\right)$

• substitutes initial conditions
• finds C

- (ii) When does the particle come to the rest first time?

1

$$\text{rest } \therefore v = 0$$

$$0 = 2 \sin\left(t - \frac{\pi}{3}\right)$$

$$t = \frac{\pi}{3} \text{ (seconds)}$$

1 mark

• correct solns.

Question 15 continued on the next page

(iii) the exact initial velocity.

$$t=0 \therefore v = 2 \sin\left(0 - \frac{\pi}{3}\right)$$

$$v = 2 \times \sin\left(-\frac{\pi}{3}\right)$$

$$v = 2 \times -\frac{\sqrt{3}}{2}$$

$$\therefore v = -\sqrt{3} \text{ m/s}$$

1 mark

• correct solns.

(iv) the maximum speed of the particle.

$$\text{speed} = |\text{velocity}|$$

→ velocity graph
is a function of

$$2 \sin\left(t - \frac{\pi}{3}\right)$$

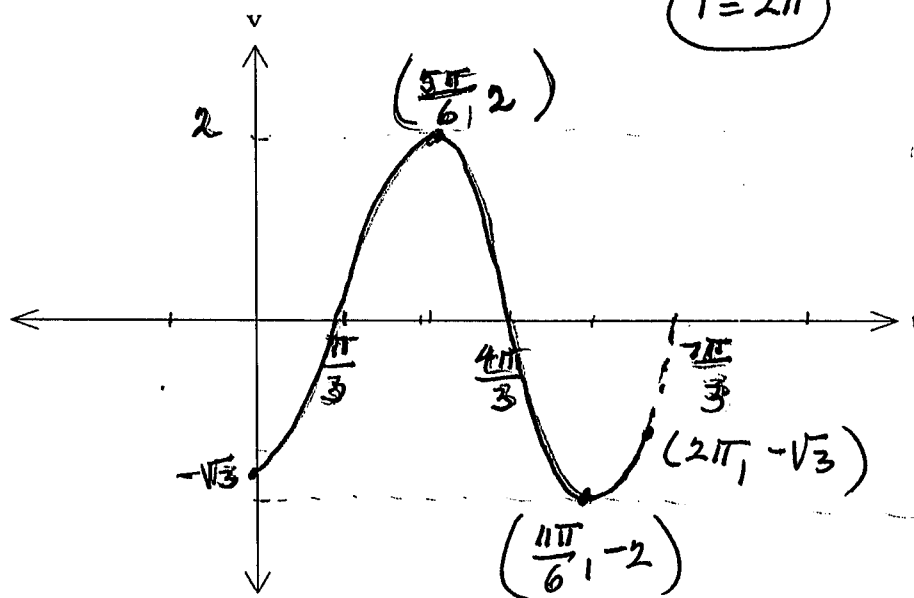
$$\therefore \text{speed}_{\text{max}} = \text{amplitude} = 2 \text{ m/s}$$

1 mark

• correct solns.

(v) Draw the graph of the velocity $v = 2 \sin\left(t - \frac{\pi}{3}\right)$ for $0 \leq t \leq 2\pi$.

Show x, y intercepts and stationary points.



$$T = 2\pi$$

3 marks

• correct graph

2 marks

• correct shape
showing $T = 2\pi$,
and initial $v = -\sqrt{3}$
and zero velocity
at $t = \frac{\pi}{3}$

1 mark

• correct shape
• initial velocity
• $v = 0$ correctly

Question 15 continued on the next page

- (vi) Using your graph above or otherwise, find the total distance travelled by the particle in the first π seconds.

2

$$d = \left| \int_0^{\pi/3} v dt \right| + \int_{\pi/3}^{\pi} v dt$$

2 marks

• correct solus

$$d = \left| \left[-2 \cos \left(t - \frac{\pi}{3} \right) \right]_0^{\pi/3} \right|$$

1 mark

• correctly puts the first section $0 \leq t \leq \frac{\pi}{3}$ into absolute value

$$+ \left[-2 \cos \left(t - \frac{\pi}{3} \right) \right]_{\pi/3}^{\pi}$$

• Finds $\int_0^{\pi} v dt = 2$

$$= \left| -2 (\cos 0 - \cos(\frac{\pi}{3})) \right|$$

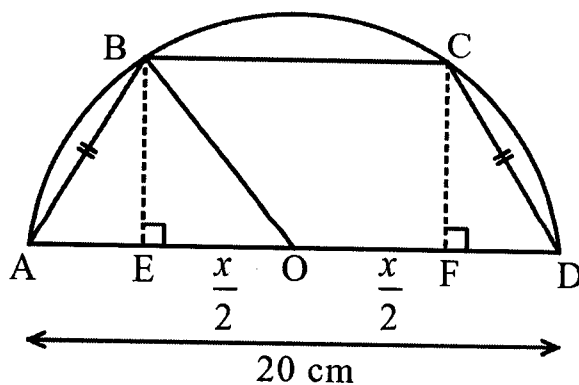
$$+ -2 \left[\cos \frac{2\pi}{3} - \cos 0 \right]$$

$$= 1 + 3 = 4 \text{ metres}$$

End of Question 15

Question 16 (7 Marks)

An isosceles trapezium $ABCD$ is drawn with its vertices on a semicircle centre O and diameter 20 cm. Perpendiculars BE and CF are drawn to meet the diameter AD as shown in the diagram below.



- (i) If $EO = \frac{x}{2}$, show that $BE = \frac{1}{2}\sqrt{400 - x^2}$.

$BO = \text{radius} = 10\text{cm}$
 in $\triangle BEO$: $BE^2 = BO^2 - EO^2$
 $BE^2 = 10^2 - \left(\frac{x}{2}\right)^2$
 $BE^2 = 100 - \frac{x^2}{4} = \frac{400 - x^2}{4}$
 $\therefore BE = \frac{1}{2}\sqrt{400 - x^2}$
 $BE > 0$

2 marks
 • correct solus.
 (1 mark)
 • applies Pythagoras's theorem correctly

- (ii) Show that the area of the trapezium $ABCD$ is given by

$$A = \frac{1}{4}(x + 20)\sqrt{400 - x^2}$$

$BC = x$
 $A = \frac{1}{2} \text{height} (BC + AD)$
 $\text{height} = BE$ (part (i))
 $\therefore A = \frac{1}{2} \times \frac{1}{2} \sqrt{400 - x^2} (x + 20)$
 $\therefore A = \frac{1}{4} (x + 20) \sqrt{400 - x^2}$

2 marks
 • correct solus.
 (1 mark)
 • applies correctly the area formula of trapezium
 • recognises $BC = x$ and height = BE

Question 16 continued on the next page

- (iii) Hence find the length of BC so that the area of the trapezium $ABCD$ is maximum.

3

find $BC = x$

(3 marks)

$$A = \frac{1}{4} (x+20) \sqrt{400-x^2}$$

• correct solus

(2 marks)

$$u = \frac{1}{4} (x+20) \quad v = (400-x^2)^{\frac{1}{2}}$$

• applies product rule correctly and

$$u' = \frac{1}{4}$$

$$v' = \frac{1}{2} x - 2x (400-x^2)^{-\frac{1}{2}} \quad \text{solves for } \frac{dA}{dx} = 0$$

$$v' = -x (400-x^2)^{-\frac{1}{2}}$$

(1 mark)

$$\frac{dA}{dx} = \frac{1}{4} (400-x^2)^{\frac{1}{2}} + \frac{1}{4} (x+20) \cdot -x (400-x^2)^{-\frac{1}{2}}$$

• finds $\frac{dA}{dx}$ correctly

• attempts to solve

$$0 = \frac{1}{4} (400-x^2)^{\frac{1}{2}} - \frac{x(x+20)}{4(400-x^2)^{\frac{1}{2}}}$$

$$\frac{dA}{dx} = 0$$

• find a solution

$$0 = \frac{(400-x^2) - x(x+20)}{4(400-x^2)^{\frac{1}{2}}}$$

$$\text{to } \frac{dA}{dx} = 0$$

justifies the maximum

$$0 = 400 - x^2 - x^2 - 20x$$

$$0 = 2x^2 + 20x - 400$$

$$0 = x^2 + 10x - 200$$

$$0 = (x+20)(x-10)$$

$$x = -20$$

$$x = 10$$

justifies Maximum

but $x > 0$

∴ impossible

End of Exam

x	9	10	11
$\frac{dA}{dx}$	0.81	0	-0.927
	(+)	$x=10$	(-)
	(+)		(-)

∴ at $x = BC = 10\text{cm}$
Area is Maximum



BAULKHAM HILLS HIGH SCHOOL

HSC Assessment Task 2 2020

Mathematics Advanced

General Instructions

- Reading time – 5 minutes
- Working time – 90 minutes
- Write using black pen only
- Board-approved calculators may be used
- Show all necessary working in Questions 11-16
- Marks may be deducted for careless or badly arranged work
- Reference sheet is provided at the end.

Total marks – 59

This paper consists of TWO sections.

Section 1 – Page 2-5 (10 marks)

Questions 1-10

- Attempt questions 1-10
- Allow about 15 minutes for this section

Section II – Pages 6-19 (49 marks)

- Attempt questions 11-19
- Allow about 75 minutes for this section

Section I - 10 marks Answer on the multiple choice page

1. The derivative of $y = \sin x^\circ$ is

(A) $\frac{dy}{dx} = \cos x^\circ$

(B) $\frac{dy}{dx} = -\sin x^\circ$

(C) $\frac{dy}{dx} = \frac{\pi}{180} \cos \frac{\pi}{180} x$

(D) $\frac{dy}{dx} = -\frac{\pi}{180} \cos \frac{\pi}{180} x$

2. $\lim_{x \rightarrow 0} \frac{\sin 2x}{3x} =$

(A) $\frac{2}{3}$

(B) 0

(C) $\frac{3}{2}$

(D) 1

3. The function $y = f(x)$ is continuous for all x .

Given that $\int_0^1 f(x) dx = 6$ and $\int_0^3 f(x) dx = 5$, what is the value of $\int_1^3 f(x) dx$?

(A) -1

(B) 0

(C) 1

(D) 2

4. What is the period of the function $y = 4 \sin\left(\frac{x}{3}\right)$?

(A) 6π

(B) 4

(C) $\frac{2\pi}{3}$

(D) $\frac{1}{4}$

4. The domain for the function $y = \log_e(2x - 1)$ is:

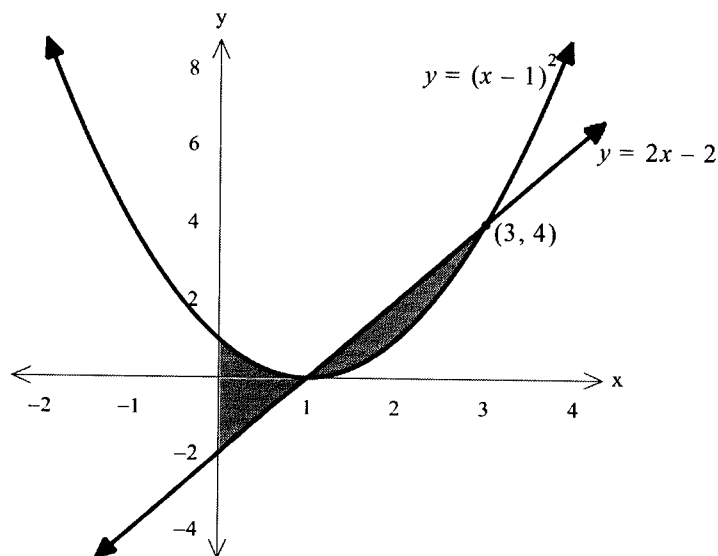
(A) All real $x, x \neq \frac{1}{2}$

(B) $x \geq \frac{1}{2}$

(C) $x > \frac{1}{2}$

(D) $x \leq \frac{1}{2}$

5. The area of the shaded region in the diagram below is given by:



(A) $\int_0^3 2x - 2 - (x - 1)^2 dx$

(B) $\int_1^3 (x - 1)^2 - 2x - 2 dx + \int_0^1 2x - 2 - (x - 1)^2 dx$

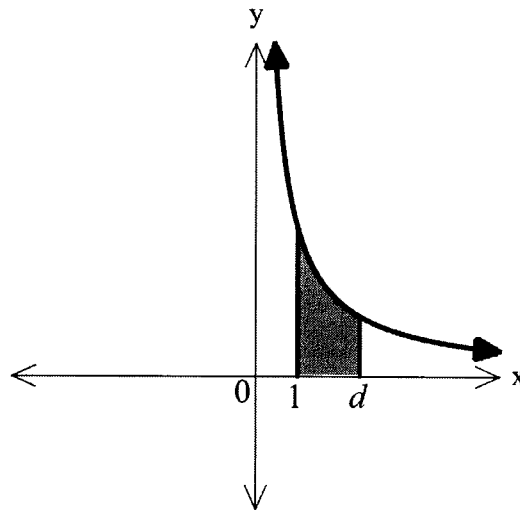
(C) $\int_0^3 2x - 2 - (x - 1)^2 dx$

(D) $\int_0^1 ((x - 1)^2 - 2x + 2) dx + \int_1^3 (2x - 2) - (x - 1)^2 dx$

7. What is $f'(x)$ if $f(x) = \ln(\tan x)$?

- (A) $\frac{1}{\sec^2 x}$
- (B) $\frac{\sec x}{\tan x}$
- (C) $\sec x \cdot \operatorname{cosec} x$
- (D) $\cot x$

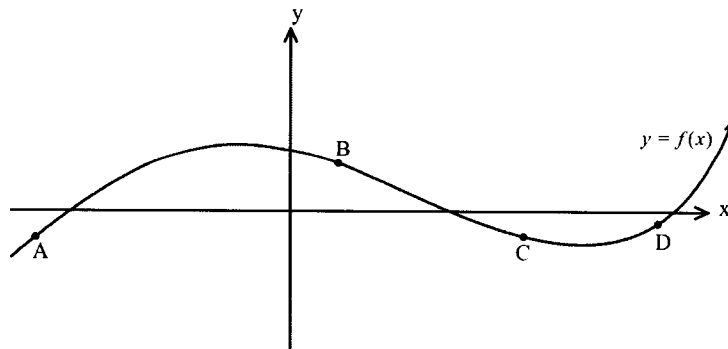
8. The diagram shows the area under the curve $y = \frac{2}{x}$ from $x = 1$ to $x = d$.



What value of d makes the shaded area equal to 2 square units?

- (A) e
- (B) $e + 1$
- (C) $2e$
- (D) e^2

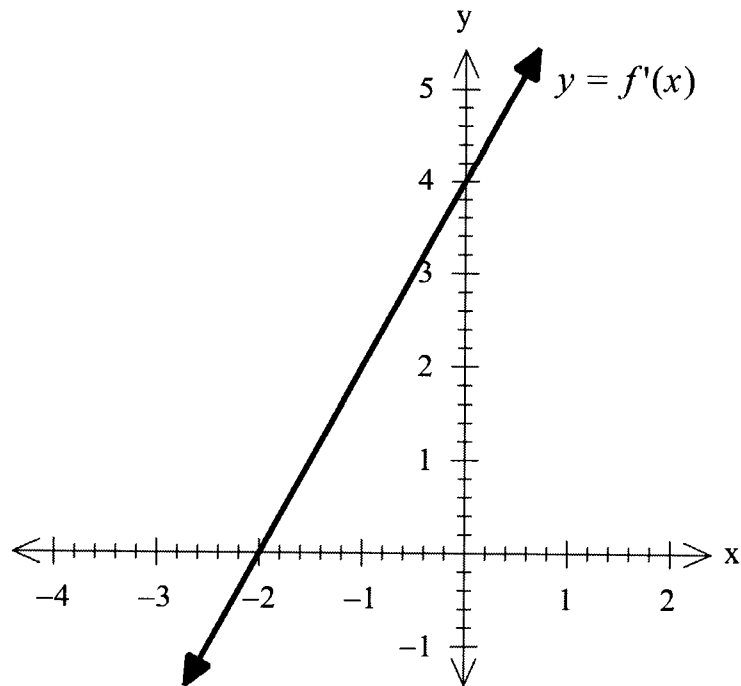
9. The diagram shows the points A, B, C and D on the graph $y = f(x)$



At which point is $f(x) < 0$, $f'(x) > 0$ and $f''(x) < 0$?

- (A) A
- (B) B
- (C) C
- (D) D

10. The graph of $y = f'(x)$ is given below.



The curve $y = f(x)$ has a minimum value of 6. What is the equation of the curve ?

- (A) $y = x^2 - 4x + 2$
- (B) $y = x^2 - 4x + 10$
- (C) $y = x^2 + 4x + 2$
- (D) $y = x^2 + 4x + 10$

END OF SECTION I

Section II- Extended Response

Attempt Questions 11-16.

Allow about 75 minutes for this section.

Question 11(12 Marks)

(a) Differentiate the following

(i) $y = \log_e(2x - 1)^4$ 2

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(ii) $y = \frac{x}{\sin^2 x}$ 2

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(iii) $y = 3^x$ 1

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Question 11 continued on the next page

b) Integrate the following

$$\int 3e^{5x+1} - \frac{1}{x^3} dx$$

2

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c) Evaluate $\int_0^{\frac{\pi}{8}} \sec^2 2x dx$

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Question 11 continued on the next page

Question 12 (7 Marks)

(a)

(i) Differentiate $\cos 3x^3$

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(ii) Hence find $\int x^2 \sin 3x^3 dx$

2

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Question 12 continued on the next page

(b) Solve the equation given by $2\ln x = \ln(3 - 2x)$

3

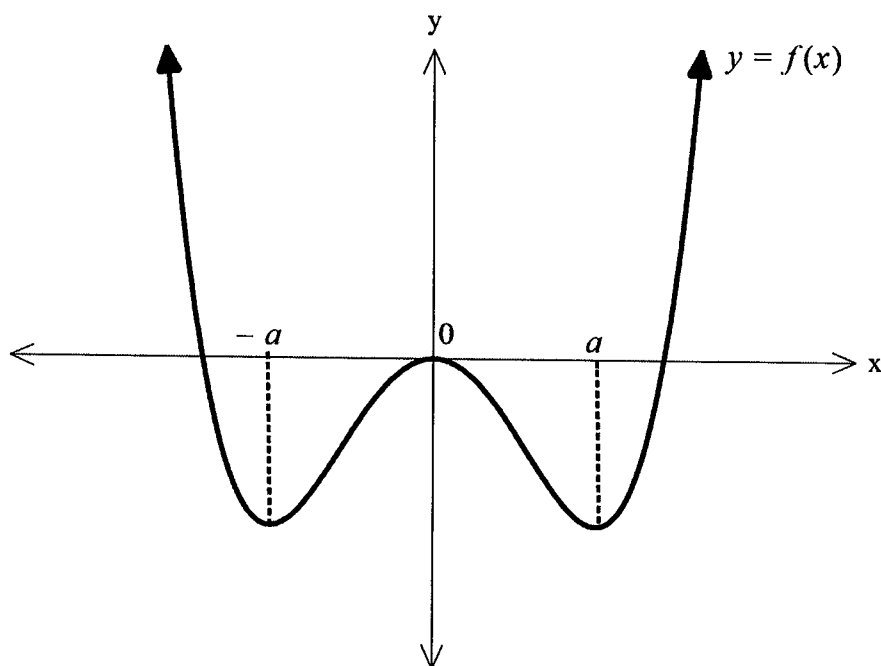
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End of Question 12

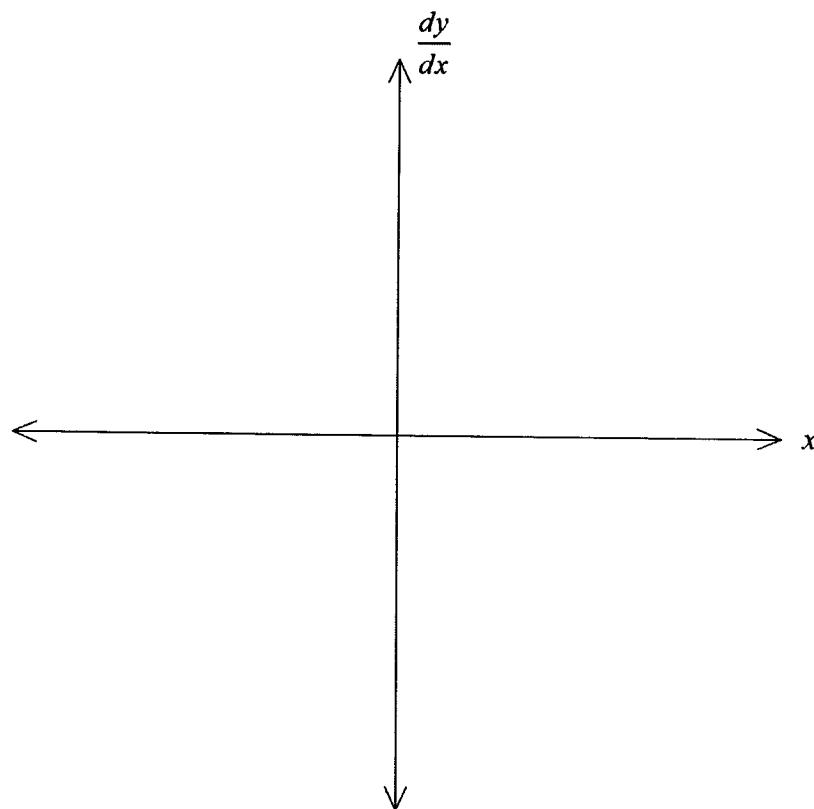
Question 13 (5 Marks)

(a) Given the graph of $y = f(x)$ below

2



Sketch a possible graph of $\frac{dy}{dx}$ on the axes provided.



Question 13 continued on the next page

(b) Water is flowing out of a pool at a rate given by $R = 4t - t^2$ cubic metres/minute. 3

If the volume of water in the pool is initially 10 cubic metres, find the amount of the water in the pool after 4 minutes.

[illegible]

End of Question 13

Question 14 (8 marks)

For the curve given by $f(x) = 2xe^{-x}$

- (i) Show that $f'(x) = 2e^{-x}(1 - x)$. 1

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- (ii) Hence or otherwise find any stationary point(s) and determine their nature. 2

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- (iii) Determine $\lim_{x \rightarrow \infty} f(x)$. 1

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Question 14 continued on the next page

(iv) Find any possible point of inflection.

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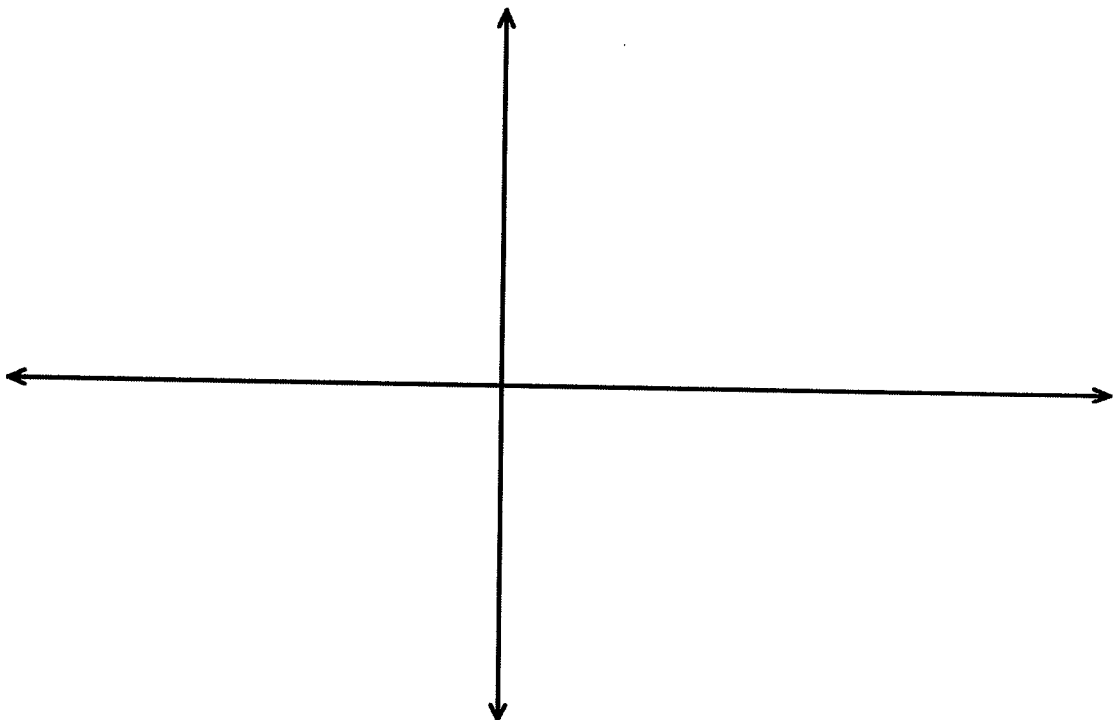
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(v) Hence sketch the curve.

2



End of Question 14

Question 15 (10 marks)

The velocity of a particle is given by $v = 2 \sin\left(t - \frac{\pi}{3}\right)$ metres per second.

Find:

- (i) the equation of the displacement x of the particle as a function of time if the particle is initially at the origin. **2**

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- (ii) when the particle comes to rest for first time. **1**

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Question 15 continued on the next page

(iii) the exact initial velocity.

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(iv) the maximum speed of the particle.

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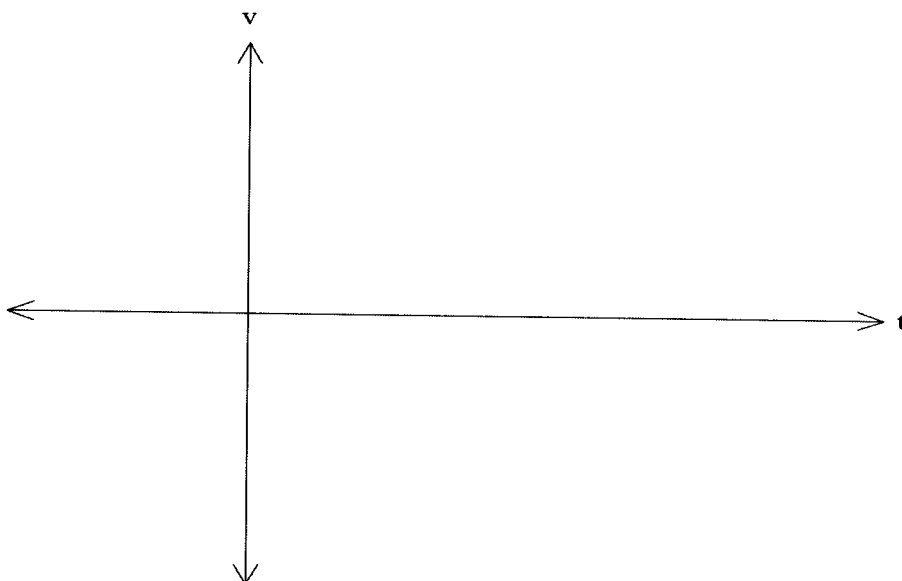
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(v) Draw the graph of the velocity $v = 2\sin\left(t - \frac{\pi}{3}\right)$ for $0 \leq t \leq 2\pi$.

3

Show all intercepts and stationary points.



Question 15 continued on the next page

- (vi) Using your graph above or otherwise, find the total distance travelled by the particle in the first π seconds.

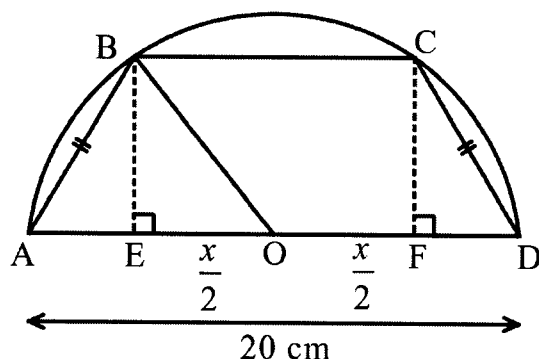
2

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End of Question 15

Question 16 (7 Marks)

An isosceles trapezium $ABCD$ is drawn with its vertices on a semicircle centre O and diameter 20 cm. Perpendiculars BE and CF are drawn to meet the diameter AD as shown in the diagram below.



- (i) If $EO = OF = \frac{x}{2}$, show that $BE = \frac{1}{2}\sqrt{400 - x^2}$. 2

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- (ii) Show that the area of the trapezium $ABCD$ is given by

$$A = \frac{1}{4}(x + 20)\sqrt{400 - x^2} . \quad 2$$

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Question 16 continued on the next page

- (iii) Hence find the length of BC so that the area of the trapezium $ABCD$ is maximum.

3

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End of Exam



BAULKHAM HILLS HIGH SCHOOL

YEAR 12 – MATHEMATICS

NESA# _____

Mathematics

Section I – Multiple Choice

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
correct

- Start Here** →
- | | | | | | | | |
|----------------------------|-------------------------|-------------------------|-------------------------|-----------------------------|-------------------------|-------------------------|-------------------------|
| 1. A ☐ | B ☐ | C ☐ | D ☐ | 6. A ☐ | B ☐ | C ☐ | D ☐ |
| 2. A ☐ | B ☐ | C ☐ | D ☐ | 7. A ☐ | B ☐ | C ☐ | D ☐ |
| 3. A ☐ | B ☐ | C ☐ | D ☐ | 8. A ☐ | B ☐ | C ☐ | D ☐ |
| 4. A ☐ | B ☐ | C ☐ | D ☐ | 9. A ☐ | B ☐ | C ☐ | D ☐ |
| 5. A ☐ | B ☐ | C ☐ | D ☐ | 10. A ☐ | B ☐ | C ☐ | D ☐ |